



Artificial intelligence and machine learning-driven advancements in gastrointestinal cancer: Paving the way for precision medicine

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Abstract

Gastrointestinal (GI) cancers remain a leading cause of cancer-related morbidity and mortality worldwide. Artificial intelligence (AI), particularly machine learning and deep learning (DL), has shown promise in enhancing cancer detection, diagnosis, and prognostication. A narrative review of literature published from January 2015 to March 2025 was conducted using PubMed, Web of Science, and Scopus. Search terms included "gastrointestinal cancer", "artificial intelligence", "machine learning", "deep learning", "radiomics", "multimodal detection" and "predictive modeling". Studies were included if they focused on clinically relevant AI applications in GI oncology. AI algorithms for GI cancer detection have achieved high performance across imaging modalities, with endoscopic DL systems reporting accuracies of 85%-97% for polyp detection and segmentation. Radiomics-based models have predicted molecular biomarkers such as programmed cell death ligand 2 expression with area under the curves up to 0.92. Large language models applied to radiology reports demonstrated diagnostic accuracy comparable to junior radiologists (78.9% vs 80.0%), though without incremental value when combined with human interpretation. Multimodal AI approaches integrating imaging, pathology, and clinical data show emerging potential for

precision oncology. AI in GI oncology has reached clinically relevant accuracy levels in multiple diagnostic tasks, with multimodal approaches and predictive biomarker modeling offering new opportunities for personalized care. However, broader validation, integration into clinical workflows, and attention to ethical, legal, and social implications remain critical for widespread adoption.

Key Words: Artificial intelligence; Gastrointestinal cancer; Precision medicine; Multimodal detection; Machine learning

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Core Tip: Gastrointestinal (GI) cancers remain a major global health burden, demanding better early detection and personalized treatments. Recent artificial intelligence (AI) advances enable precision oncology by integrating diverse data from endoscopic images to genomic profiles. AI-driven tools enhance polyp detection, tumor grading, and multi-omics analysis for tailored therapies. Despite challenges in standardization and clinical adoption, these innovations promise to reduce diagnostic disparities and improve outcomes in GI cancer care.

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INTRODUCTION

Gastrointestinal (GI) cancers-including gastric, colorectal, pancreatic, esophageal, and liver cancers-represent approximately 26% of global cancer cases and are responsible for 35% of cancer-related deaths[1-3]. The considerable anatomical diversity, molecular heterogeneity, and the tendency for late-stage diagnosis make these cancers particularly challenging to manage. Artificial intelligence (AI) is addressing these challenges by enabling early detection, analyzing subtle endoscopic and imaging features that may be imperceptible to humans, and decoding spatial tumor microenvironment characteristics from standard hematoxylin-eosin (HE) slides[4]. Additionally, AI has demonstrated utility in predicting immunotherapy response [area under the curve (AUC) = 0.86] and identifying effective drug combinations. Notably, recent advances such as the GI AI diagnostic system (GRAIDS) have achieved a diagnostic accuracy of 95.5% for upper GI cancers, surpassing the performance of junior endoscopists[5]. Now, AI has emerged as a transformative force in oncology research, revolutionizing approaches to cancer detection, diagnosis, and treatment[6]. The integration of AI technologies, particularly machine learning (ML) and deep learning (DL), has led to significant advancements across the spectrum of oncology research (as illustrated in Figure 1), from basic science to clinical applications[7].

AI algorithms are analyzing large-scale genomic, proteomic, and imaging datasets to identify novel biomarkers for cancer diagnosis, prognosis, and treatment response prediction[8]. These biomarkers are crucial for advancing personalized medicine approaches in oncology. AI tools, such as AlphaFold, have revolutionized protein structure prediction, enabling researchers to uncover novel therapeutic targets and understand complex protein-protein interactions in cancer pathways[9]. Along with that, other advanced AI algorithms are being employed to analyze multiomic and spatial pathology data, providing unprecedented insights into the intricate molecular landscapes within tumors[10]. Such AI-driven approaches for creating digital twins and other synthetic data are accelerating the design and execution of clinical trials, addressing challenges related to data scarcity and privacy concerns[11]. Moreover, AI is significantly accelerating the drug discovery process by predicting drug-target interactions, optimizing lead compounds, and identifying potential repurposing opportunities for existing drugs[12]. This has led to a more efficient and cost-effective drug development pipeline in oncology. AI-based predictive platforms are being developed to forecast treatment outcomes and guide therapy selection. Between 2018 and 2022 (Table 1 shows complete evolution history), 28% of foods and drugs act-approved first-in-class drugs were indicated for cancer treatment, highlighting the importance of efficient trial design in oncology[13].

AI in medical research encompasses various computational approaches for learning from data. Supervised learning involves training models on labeled datasets, where both input features and corresponding outcomes (e.g., imaging features and confirmed diagnoses) are known, enabling accurate prediction of similar outcomes for new cases[14]. Unsupervised learning, by contrast, identifies hidden patterns or clusters in unlabeled data useful for discovering novel GI cancer subtypes or stratifying patient populations[15]. DL architectures have been particularly transformative in medical imaging. Convolutional neural networks (CNNs) excel in detecting subtle abnormalities in computed tomography (CT), magnetic resonance imaging (MRI), and endoscopic images by automatically learning hierarchical features, while recurrent neural networks process sequential medical data, such as time-series biomarker profiles.

A critical component of AI model development is featuring engineering, where domain knowledge is applied to extract, select, or transform relevant variables from complex datasets. In medical data analysis, this may involve deriving texture features from radiomics or extracting quantitative biomarkers from pathology slides to enhance model accuracy and interpretability[16]. The methodological frameworks used in GI cancer research share similarities with AI applic-

Table 1 Historical evolution of artificial intelligence applications in gastrointestinal cancer management[5,46-48,123,173-175]

Year	Ref.	Country	Focus area	AI technique used	Dataset/study design	Key findings	Clinical impact/advancement
2015	Miyaki <i>et al</i> [173]	Japan	Early gastric cancer	SVM	100 cases (retrospective)	Achieved 84.6% accuracy in distinguishing EGC using blue-laser imaging	Demonstrated feasibility of ML in endoscopic analysis
2017	Hirasawa <i>et al</i> [46]	Japan	Gastric cancer detection	CNN (SSD architecture)	13584 images (retrospective)	92.2% sensitivity in detecting gastric cancer from endoscopic images	Validated AI's potential for real-time lesion detection
2018	Luo <i>et al</i> [47]	China	Upper GI cancer screening	GRAIDS (CNN-based)	844424 cases (prospective)	95.5% diagnostic accuracy for upper GI cancers in real-time endoscopy	First real-time AI system for mass screening
2019	Zhu <i>et al</i> [48]	China	Invasion depth prediction	CNN (ResNet50)	993 images (retrospective)	89.16% accuracy in predicting gastric cancer invasion depth <i>via</i> endoscopy	Enhanced preoperative staging accuracy
2020	Nagao <i>et al</i> [174]	Japan	Metastasis prediction	ResNet50	16557 images (retrospective)	94.5% accuracy in identifying lymph node metastasis from CT images	Improved non-invasive metastasis assessment
2021	Hu <i>et al</i> [175]	China	Tumor margin delineation	VGG-16	694 images (retrospective)	82.7% accuracy in differentiating EGC margins under magnifying endoscopy	Supported precise endoscopic resection planning
2022	Wu <i>et al</i> [125]	China	Survival prediction	VGG-16, ResNet-50	100 videos (prospective)	78.57% accuracy in predicting survival and invasion depth in real-time EGD	Reduced diagnostic time by 90% compared to experts
2024	Mukherjee <i>et al</i> [5]	Global	Comprehensive review	ML/DL models	Meta-analysis of 50 + studies	Highlighted AI's role in early detection (AUC: 0.86-0.94 across modalities)	Synthesized evidence for AI-driven personalized care

AI: Artificial intelligence; GI: Gastrointestinal; ML: Machine learning; DL: Deep learning; SVM: Support vector machine; CNN: Convolutional neural network; SSD: Single-shot detector; GRAIDS: Gastrointestinal artificial intelligence diagnostic system; VGG-16: Visual geometry group 16; EGC: Early gastric cancer; CT: Computed tomography; EGD: Esophagogastroduodenoscopy; AUC: Area under the curve.

ations in other biomedical and environmental domains, including ML modeling integrated with experimental validation and optimization techniques. The schematic representation of general artificial intelligence-based workflow in gastrointestinal cancer research is given in [Figure 1](#). Examples include deep eutectic solvent-modified polyvinyl alcohol/chitosan thin film membranes for dye adsorption[17]. These multidisciplinary applications highlight the versatility of AI in processing complex datasets and optimizing predictive performance, underscoring its potential in advancing GI cancer detection, diagnosis, and treatment planning.

AI IN GI CANCER DETECTION AND DIAGNOSIS

AI has been integrated into all phases of GI cancer diagnosis, including endoscopy, histopathology, and a range of radiological imaging modalities such as CT and MRI[18]. AI has improved both the accuracy and efficiency of gastric cancer assessment by precisely analyzing data from traditional imaging techniques, overcoming challenges such as variable lighting and reflection artifacts that can affect endoscopic image quality[19]. The AI system ENDOANGEL-ED was developed to train, validate, and test the detection of focal lesions in human endoscopic images. It demonstrated higher accuracy than endoscopists in both internal validation (81.10% *vs* 70.61%) and external validation (88.24% *vs* 78.49%)[20]. Further, Li *et al*[21] developed a DL system capable of classifying four types of gastric lesions (advanced cancer, early gastric cancer, dysplasia, and non-tumor) with an accuracy of 89.7%. AI also significantly improved the diagnostic specificity of intermediate physicians (59.79% *vs* 52.62%), while maintaining comparable sensitivity, highlighting AI's ability to enhance the diagnostic performance of non-expert clinicians in identifying gastric tumors[21]. Niikura *et al*[22] analyzed 500 patient images, including 100 gastric cancer cases, and found that the AI group achieved a higher diagnostic accuracy than experts (100% *vs* 94.12%) and superior specificity (89.87% *vs* 81.17%).

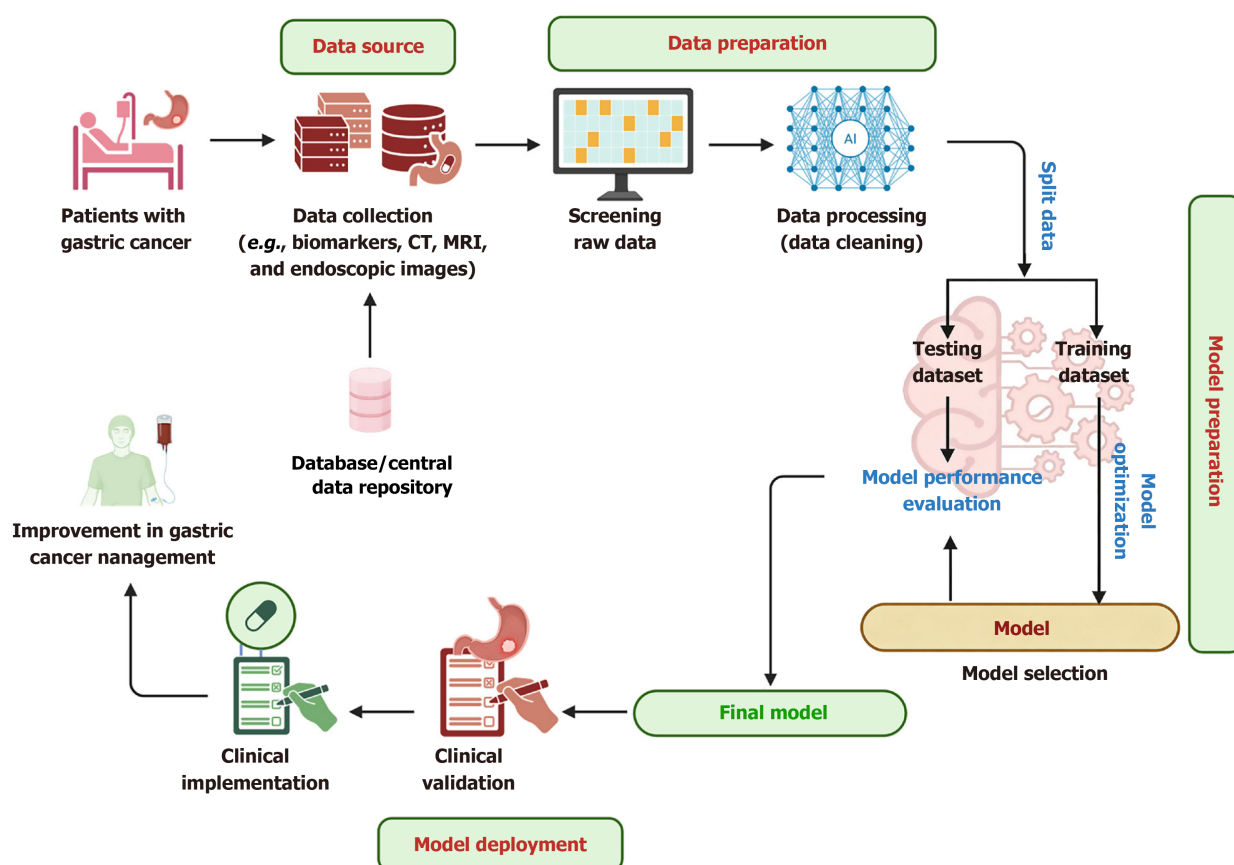


Figure 1 Schematic representation of a general artificial intelligence-based workflow in gastrointestinal cancer research. AI: Artificial intelligence; CT: Computed tomography; MRI: Magnetic resonance imaging.

Furthermore, an improved mask R (IMR-CNN) model to identify and segment gastric cancer lesions in gastroscopic images by identifying and verifying early gastric cancer images, with precision, recall, accuracy, specificity, and F1-score values of 92.9%, 95.3%, 93.9%, 92.5%, and 94.1%, respectively[23]. Du *et al*[24] proposed a three-branch automatic segmentation framework based on co-spatial attention and channel attention (CSA-CA-TB-ResUnet), which achieved a Jaccard similarity index of 84.54%, a threshold Jaccard index of 81.73%, a Dice similarity coefficient of 91.08%, and a pixel-level accuracy of 91.18%, proving that the correlation information between gastroscopic images can improve the accurate segmentation of early gastric cancer lesions[24]. Sun *et al*[25] proposed a new network for gastric lesion segmentation using generative adversarial training, and the experimental results showed that the dice value, accuracy, and recall rate were 86.6%, 91.9%, and 87.3%, respectively, which were significantly better than the existing models. A 2025 study found that a two-step ChatGPT-4o approach achieved diagnostic accuracy for focal liver lesions comparable to junior radiologists but slightly lower than middle-level radiologists, without adding incremental value to radiologist interpretations[26].

Endoscopy and pathological biopsy remain the primary diagnostic methods for GI tumors, with pathology considered the gold standard[27]. Nevertheless, the field faces significant challenges, including a growing shortage of pathologists-particularly in remote or resource-limited settings-and the impact of prolonged, high-intensity workloads, which can reduce diagnostic accuracy due to fatigue and variability in clinical experience among physicians[28]. These factors contribute to increased uncertainty and potential for diagnostic errors. The integration of AI technology has brought transformative potential to this traditional field. AI enables efficient and automated analysis of whole-slide digital images (WSI) by using advanced data processing and DL algorithms[29]. This technology can accurately identify lesion types and assess lesion severity, providing robust auxiliary support-especially for less experienced clinicians. ML algorithms have demonstrated significant potential in early cancer detection and diagnosis, offering unprecedented accuracy and efficiency across various domains of oncology[30]. A study utilizing ML techniques achieved 97% accuracy in diagnosing two common types of lung cancer through tissue sample analysis[31]. AI-powered chatbots and virtual assistants are being developed to help individuals monitor and recognize early cancer symptoms, prompting timely medical attention [32]. The cancer data-driven detection programme, launched in 2025, aims to develop AI tools that analyze vast quantities of data to predict individual cancer risk throughout a person's lifetime[33]. AI is also advancing the molecular characterization of tumors. The Clinical Histopathology Imaging Evaluation Foundation (CHIEF) model, developed in 2024, leverages AI to extract diverse pathology representations for systematic cancer evaluation[21]. CHIEF has shown success in cancer cell detection, tumor origin identification, and molecular profile characterization, outperforming state-of-the-art DL methods by up to 36.1%[33].

CNN based systems significantly improve early lesion detection in esophageal and stomach cancers by leveraging advanced image recognition capabilities to analyze endoscopic images with high accuracy and speed[34]. These AI models are trained on large datasets of annotated endoscopic images, enabling them to identify subtle mucosal changes and early neoplastic lesions that may be missed by human endoscopists[35].

In clinical studies and meta-analyses, CNN systems have demonstrated sensitivities and specificities exceeding 90% for early esophageal and gastric cancer detection with pooled AUC values as high as 0.95-0.98, outperforming or matching expert endoscopists[36]. CNNs also excel in real-time analysis, providing immediate feedback during endoscopic procedures and reducing missed diagnoses. In randomized clinical trials CNN-assisted endoscopy doubled the detection rate of high-risk esophageal lesions compared to standard practice, while maintaining high specificity and accuracy[37]. Achieve 92% sensitivity for esophageal/gastric cancers in a meta-analysis of 14 studies, detecting subtle mucosal changes invisible to human endoscopists. AI systems estimate tumor invasion depth with 89% histologic concordance, outperforming expert visual assessment for resection plannings. Moreover, the GRAIDS reduces missed diagnoses by 41% through real-time anomaly detection during endoscopy[38].

AI spatial analysis of HE slides reveals stromal tumor-infiltrating lymphocytes density $> 1000/\text{mm}^2$ predicts 76% 5-year disease-free survival *vs* 34% in tumor-infiltrating lymphocytes-low colon cancer patients (hazard ratio = 0.142, $P < 0.0001$)[39]. The CHIEF model deciphers microsatellite instability (MSI) status from routine histology with AUC = 0.91 (95% confidence interval: 0.88-0.94), matching immunohistochemistry performance while eliminating additional testing. The integration of CNNs into endoscopic workflows not only enhanced diagnostic accuracy but also standardizes assessments, compensates for interobserver variabilities, and accelerates the diagnostic process by analyzing images at speeds exceeding 30 frames per second. Overall, CNN-based systems represent a powerful tool for improving early detection and clinical outcomes in esophageal and stomach cancers. The tumor microenvironment is a dynamic ecosystem comprising tumor cells, immune cells, stromal cells, and extracellular matrix components[40]. Mask R-CNN segments and classifies immune cells, in HE slides with 95% accuracy, distinguishing M1 (pro-inflammatory) from M2 (anti-inflammatory) phenotypes based on morphology[41]. On the other hand, graph neural networks map spatial interactions between cytotoxic T-cell proximity to tumor cells, predicting immunotherapy response (AUC = 0.86 in non-small cell lung cancer)[42]. Another tool, SiQ-3D is a DL pipeline that tracks 3-D cell-cell interactions in patient-derived organoids, revealing natural killer-cell mediated tumor killing dynamics[43]. This approach addresses longstanding limitations of traditional methods like immunoscore, which relies on labor-intensive immunohistochemistry. The AI's ability to stratify risk groups using routine HE slides enhances clinical feasibility, offering pathologists an objective, scalable tool for precision prognostics[44]. Concurrently, systematic reviews highlight AI's broader role in decoding tumor microenvironments, achieving up to 97.7% accuracy in predicting features like tumor budding and immune cell spatial patterns[45].

AI in GI molecular pathways analysis

The integration of AI and ML into the analysis of molecular pathways in GI diseases has significantly transformed the landscape of precision medicine[46-49]. This is particularly relevant for complex and multifactorial conditions such as inflammatory bowel diseases (IBD), irritable bowel syndrome, and GI malignancies, including colorectal cancer (CRC) and gastric adenocarcinoma. It is now possible to interpret vast and multidimensional datasets including genomic, transcriptomic, proteomic, metabolomic, and histopathological information with unprecedented precision[50]. These AI-driven insights enable researchers and clinicians to uncover previously undetectable disease mechanisms, improve prognostic accuracy, and design more effective, patient-specific therapeutic strategies[51]. A key example is the chronic inflammation characteristic of IBD, which is driven by pro-inflammatory signaling cascades such as nuclear factor kappa-B, tumor necrosis factor (TNF)- α , and the interleukin (IL)-6/IL-23/IL-17 axis. These pathways not only mediate mucosal injury but also create a pro-tumorigenic environment conducive to cancer development[52]. Prolonged activation of these inflammatory circuits contributes to sustained oxidative stress, DNA damage, and genomic instability, collectively promoting the pathogenesis of colitis-associated CRC a known complication of chronic ulcerative colitis and Crohn's disease[53]. AI-based analytical platforms have shown great potential in forecasting disease trajectory and responsiveness to specific treatments, particularly biologic agents such as anti-TNF therapies[54].

Among the various oncogenic pathways implicated in CRC, the Wnt/ β -catenin signaling cascade is one of the most frequently dysregulated, with more than 90% of sporadic CRCs exhibiting mutations in key regulatory genes like APC or CTNNB1[55]. ML models trained on whole-exome sequencing data can predict the aggressiveness of Wnt pathway mutations, enhancing risk stratification[56]. The interaction between the mitogen-activated protein kinase/extracellular regulated protein kinases and phosphatidylinositol 3-kinase/protein kinase B/mammalian target of rapamycin signaling axes forms a central regulatory hub in GI cancers[57]. In CRC, activating mutations in KRAS and BRAF genes not only fuel tumorigenesis but also underlie resistance to therapies targeting the epidermal growth factor receptor (EGFR)[58]. AI has been instrumental in dissecting this molecular complexity, with neural networks trained on integrated multi-omics datasets accurately predicting resistance to targeted treatments[59]. In addition to host signaling pathways, the human gut microbiome has emerged as a key player in modulating GI cancer susceptibility. AI tools, especially DL frameworks, are well-suited to analyze the vast and heterogeneous data from metagenomic sequencing. These models have successfully identified microbial signatures that serve as early indicators of pre-malignant or dysplastic lesions, thus offering promising avenues for early intervention[60]. Detailed pathway depicted in Figure 2.

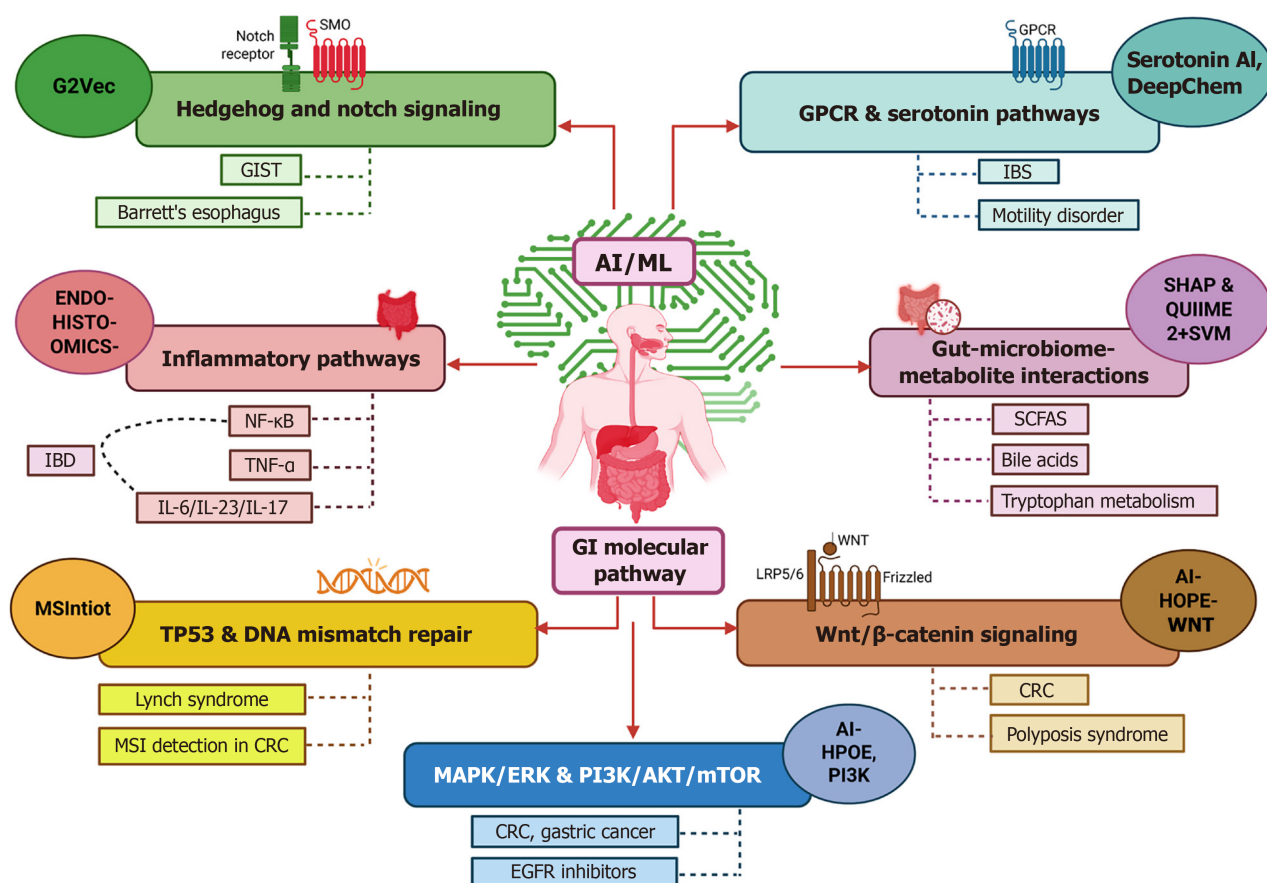


Figure 2 Artificial intelligence-driven analysis of molecular pathways in gastrointestinal cancers and associated disorders. This figure illustrates how artificial intelligence (AI) helps decode key molecular pathways involved in gastrointestinal diseases. Tools like G2Vec (a graph-based AI) are used for pathway analysis, while Hedgehog and Notch signaling pathways are linked to conditions such as Barrett's esophagus and gastrointestinal stromal tumors. ENDO-HISTO-OMICS integrates endoscopic, histological, and multi-omics data through AI to enhance diagnosis. Inflammatory pathways like nuclear factor kappa-B, tumor necrosis factor- α , and the interleukin (IL)-6/IL-23/IL-17 are evaluated in inflammatory bowel disease. AI also detects *TP53* mutations and DNA mismatch repair defects, which are associated with Lynch syndrome and microsatellite instability in colorectal cancer (CRC). Deep learning tools such as DeepChem are applied to analyze G protein-coupled receptor and serotonin pathways, which play roles in irritable bowel syndrome and gut motility disorders. Models like QUIIME 2, paired with support vector machines, help explore interactions between the gut microbiome and metabolites such as short-chain fatty acids, bile acids, and tryptophan derivatives. AI also informs responses to therapies by analyzing key signaling pathways including Wnt/ β -catenin and mitogen-activated protein kinase/extracellular regulated protein kinases/phosphatidylinositol 3-kinase/protein kinase B/mammalian target of rapamycin in CRC and gastric cancers. SMO: Site management organization; GIST: Gastrointestinal stromal tumor; AI: Artificial intelligence; GPCR: G protein-coupled receptor; IBS: Irritable bowel syndrome; ML: Machine learning; IBD: Inflammatory bowel disease; NF- κ B: Nuclear factor kappa-B; TNF: Tumor necrosis factor; IL: Interleukin; SHAP: SHapley Additive exPlanations; SVM: Support vector machine; SCFAS: Short chain fatty acid; GI: Gastrointestinal; MSI: Microsatellite Instability; CRC: Colorectal cancer; MAPK: Mitogen-activated protein kinase; ERK: Extracellular regulated protein kinases; PI3K: Phosphatidylinositol 3-kinase; AKT: Protein kinase B; mTOR: Mammalian target of rapamycin; EGFR: Epidermal growth factor receptor.

AI IN CLINICAL DECISION SUPPORT

AI-based predictive platforms are being developed to forecast treatment outcomes and guide therapy or treatment selection in GI cancers. Recent meta-analyses demonstrate that AI networks analyzing genetic mutation profiles can predict immunotherapy responses in GI cancers with impressive performance metrics. These sophisticated models integrate complex genomic signatures with clinical parameters to stratify patients and guide precision therapy. Researchers have trained ML algorithms to predict immunotherapy response in melanoma patients[61]. Mayo clinic researchers have developed a 32-gene signature that serves as both a prognostic and predictive biomarker for gastric cancer patients. This AI-derived signature enables a more granular stratification beyond traditional staging, allowing clinicians to identify patients likely to benefit from specific therapeutic approaches and guiding decisions about surgical intervention, chemotherapy timing, and immunotherapy candidacy. The NTriPath algorithm further enhances this approach by integrating somatic mutation profiles with gene-to-gene interaction networks to identify cancer-associated molecular pathways with prognostic significance. AI tools are also being developed to identify patients with lung or gastroesophageal cancers who are likely to benefit from immunotherapy[62]. The multimodal transformer with unified mask modeling AI model developed at Stanford Medicine combines data from medical images with text to predict cancer prognoses and treatment responses[63]. This model outperformed standard methods in predicting prognoses for diverse cancer types and identifying patients likely to benefit from specific treatments. AI is also accelerating the drug discovery process by predicting drug-target interactions, optimizing lead compounds, and identifying potential repurposing

opportunities for existing drugs. This has led to a more efficient and cost-effective drug development pipeline in oncology[64].

Personalized treatment planning and risk assessment

GI cancer management increasingly relies on AI-powered clinical decision support tools that synthesize multimodal data. The GastricAITool exemplifies this approach, combining ML algorithms with explainable AI methods to provide interpretable diagnostic and prognostic guidance[65]. Using extreme gradient boosting for diagnosis and random survival forest for prognosis, this tool integrates up to 261 variables-including demographic, environmental, clinical, tumoral, and genetic factors-to deliver personalized treatment recommendations[66]. The integration of genetic data significantly enhances discriminatory ability, underscoring the importance of comprehensive biomarker assessment in treatment planning. Immune checkpoint inhibitor (ICI) therapy, researchers have developed the gastric mutation signature (GMS), which reliably predicts survival benefit across multiple validation cohorts[67]. This AI-derived signature not only predicts treatment outcomes but also correlates with underlying tumor biology; low-risk samples exhibit higher infiltration of cytolytic immune cells and enhance immunogenic potential, explaining their superior response to immunotherapy. Additionally, the GMS has therapeutic implications beyond prediction, as it correlates with sensitivity to specific targeted compounds like UMI-77, suggesting potential for AI-guided drug selection[68]. As for drug response prediction, AI algorithms analyze genetic data to predict individual patient responses to various cancer therapies, enabling more personalized treatment approaches[69].

Advanced applications in GI cancer management

Endoscopic submucosal dissection (ESD) is a well-established and minimally invasive technique for the resection of early GI tumors, including undifferentiated early-stage gastric cancer, offering high overall resection rates and excellent oncologic outcomes when performed by skilled operators[70]. Recent advances in AI and ML have enhanced the precision of ESD by enabling the prediction of radical resection outcomes based on detailed morphological and ecological characteristics of lesions, with reported internal validation metrics such as accuracy, precision, recall, and F1 scores exceeding 90%[71]. Additionally, risk-scoring models have demonstrated strong discriminatory performance in predicting non-curative resections, further supporting clinical decision-making. Despite these advances, ESD remains technically demanding, particularly in challenging anatomical locations, and is associated with a risk of complications. To address these challenges, simulation-based training tools and robotic-assistive technologies have been developed. A robotic-assisted or trainee-performed ESD in difficult locations has resulted in significantly shorter operative times compared to conventional dissection, without compromising safety or completeness of resection[72]. The introduction of assistive robotic arms has also been shown to reduce the incidence of complications, particularly among less experienced operators, thereby improving overall surgical safety.

In parallel, the era of precision surgery for gastric cancer is characterized by the integration of advanced imaging, genetic testing, and minimally invasive surgical techniques tailored to individual patient profiles. Technologies such as real-time navigation systems and intraoperative imaging help ensure complete tumor resection while minimizing harm to healthy tissues, thus enhancing postoperative recovery and quality of life[73]. The adoption of these precision approaches, supported by AI and robotics, signifies a new era in gastric cancer treatment defined by increased efficiency, safety, and individualized patient care. AI systems are transforming GI cancer staging and grading through automated tissue analysis and multimodal data integration. These platforms combine imaging features, molecular profiles, and clinical parameters to provide standardized, reproducible assessment with high concordance to expert evaluation. In gastric cancer specifically, AI algorithms assist with tumor-node-metastasis (TNM) staging and molecular subtype classification, enhancing the precision of pathological evaluation while reducing inter-observer variability[74]. Beyond conventional treatment selection, AI accelerates drug discovery and repurposing for GI malignancies. Computational approaches identify novel therapeutic targets and predict drug efficacy based on molecular signatures, significantly shortening development timelines. AI analysis of drug-target interactions has identified repurposing opportunities for existing compounds against treatment-resistant gastric and CRCs, offering new options for patients with limited therapeutic alternatives[75].

Radiotherapy planning and optimization

Radiotherapy (RT) is an important component of cancer care, in which the whole process starts with a series of visits to radiation oncology (RO) clinic, culminating in the final diagnosis, staging, and prognostication after which a radiation treatment process can be categorized into imaging, target and organs at risks segmentation, treatment plan generation, onboard imaging, treatment delivery and quality assurance checks. The advancements of intensity-modulated RT and volumetric modulated arc therapy that offer exceptionally conformal RT delivery increased manifold the intricacy and complexity of RT planning. High-precision treatment procedures such as stereotactic body ablative RT often consumes hours or even days of human effort for planning.

ML techniques, including DL approaches, have dealt with intra-and intrafraction patient and organ motion during RT treatment delivery to aid tumor gating and motion tracking[76]. Frameworks have been built using neural networks trained on collected patient breathing data to predict the breathing pattern while delivering RT. ML has been used to aid motion tracking by assisting in the detection of the tumor (marker less tracking) or surrogate markers. ML has been used to help avert setup errors and patient safety hazards by tracking the treatment room components and the patient's body in real time using three dimensional cameras to fine tune a CNN for object recognition[77]. Scientists have developed a computer vision based pneumatic soft robot a curator to better estimate a patient's head pitch motion and to manipulate the patient head position based on sensed head pitch motion, thereby potentially eliminating the need for immobilization

with a thermoplastic mask. Also, algorithms can assist physicians in supervising variations during treatment course by evaluating daily setup variations and anatomic changes, for early identification of adaptive replanning requirement.

AI IN GI CANCER THERAPEUTICS

The integration of AI in cancer therapeutics has led to significant advancements in drug discovery, efficacy prediction, drug repurposing, and clinical trial optimization. Future advancements will depend on multimodal AI integrating real-world evidence, patient-derived organoids, and digital twins to simulate treatment outcomes at scale. AI is transforming cancer therapeutics from serendipitous discovery to a precision-driven science-ushering in an era of faster, safer, and more equitable patient care, by bridging computational power with biological insight (Figure 3).

Neoadjuvant chemotherapy

Neoadjuvant chemotherapy (NAC) has emerged as a transformative approach in the management of advanced gastric cancer, demonstrating significant improvements in patient prognosis. Landmark trials including MAGIC, CLASSIC, and RESOLVE have established the multifaceted benefits of NAC, which include tumor size reduction, downstaging, enhanced R0 resection rates, decreased recurrence risk, and improved overall survival rates[78]. The timely identification of suitable candidates for NAC represents a critical step in optimizing individualized treatment strategies for locally advanced gastric malignancies. With advancements in imaging technology, CT has become an invaluable tool for assessing NAC efficacy[79]. Recent innovations in AI have further enhanced this capability through the development of sophisticated DL models. One such approach is the DL CS model, which demonstrated remarkable predictive performance when trained on 1060 patients with locally advanced gastric cancer[80]. This model maintained excellent discrimination across internal validation cohorts (265 patients; AUC = 0.808) and external validation at five independent centers (AUCs: 0.755, 0.752), consistently outperforming conventional clinical assessment methods[81]. Cui *et al*[81] developed an ensemble DL radiomics nomogram using data from 719 patients across four hospitals. The statistical superiority of this ensemble model compared to conventional clinical models ($P < 0.001$) underscores its potential clinical utility. Importantly, this DL radiomics nomogram demonstrated significant association with disease-free survival, suggesting its value not only as a predictive tool but also as a prognostic indicator.

Immunotherapy

Despite the widespread application of immunotherapy across various tumor types, only a minority of patients achieve a durable clinical benefit. The overall objective response rates remain modest, and a subset of patients may experience serious immune-related adverse events[82]. This highlights the critical need for predictive biomarkers that can guide immunotherapy efficacy and prognosis. To address this, Wang *et al*[83] developed a multimodal DL radiomics framework trained on patients receiving immunotherapy. Their model achieved promising predictive performance, with an AUC of 0.791 and 0.812 in internal and external validation cohorts, respectively. Similarly, Chen *et al*[84] employed weighted gene co-expression network analysis to identify immune subtype-associated genes. Using ML, they successfully isolated optimal prognostic features across the full patient cohort. Jiang *et al*[85] proposed a non-invasive strategy to assess the tumor microenvironment by integrating radiomics and DL across a multi-institutional cohort of gastric cancer patients.

Despite the growing application of ICIs in gastric cancer, their efficacy remains limited to a small proportion of patients [86]. In this context, Huang *et al*[87] introduced the genome stabilization (GS) classifier a ML-based framework that predicts personalized immune subtypes from gene expression profiles. This classifier demonstrated strong predictive accuracy for pembrolizumab response (AUC = 0.833). Park *et al*[88] further identified ACTA2 expression as a potential predictive biomarker: In a cohort of 567 gastric cancer patients, those with low ACTA2 expression exhibited a 56% response rate to ICIs, compared to only 25% in the high-expression group. A meta-analysis by Li *et al*[89], encompassing 21 prospective phase I/II studies in gastric cancer, reported a pathological complete response rate of 21%, major pathological response rate of 41%, and an R0 resection rate of 94%. The combination of ICIs with chemoradiotherapy yielded the highest efficacy, while ICIs alone were the least effective; intermediate results were observed with ICI plus chemotherapy $\pm$ antiangiogenic agents. Notably, Han *et al*[90] developed a pathomics-based ensemble model with high accuracy in predicting ICI response. Such integrative, high-throughput tools are poised to significantly enhance the precision of immunotherapy by enabling more accurate, individualized treatment strategies.

Epstein-Barr virus-associated gastric cancer and MSI

According to the unique molecular characteristics of gastric cancer, the cancer genome map divides gastric cancer into Epstein-Barr virus infection (EBV), MSI, chromosomal instability (CIN), and GS[91]. Among them, EBV and MSI are the types that may benefit from immunotherapy, while CIN and GS are less likely to respond to immunotherapy[92]. EBV-associated gastric cancer is confirmed by expensive molecular testing (EBV-encoding RNA *in situ* hybridization). Therefore, Zheng *et al*[93] constructed the EBVNet network and fused it with pathologists. The results showed that the AUC cross-validated from the internal was 0.969, the external dataset of multiple institutes was 0.941, and the cancer genome map dataset was 0.895, indicating that the human-machine fusion significantly improved the diagnostic performance of EBVNet and pathologists. Vuong *et al*[94] trained a DL algorithm to study 137184 image patches from 16 tissue microarrays (TMAs) (708 tissue cores), 24 WSIs, and 286 gastric cancer WSIs. The classifier can classify EBV-gastric cancer image plaques of TMAs and WSI with an accuracy of 94.70%, a recall of 0.936, an accuracy of 0.938, an F1 score of 0.937, and a κ coefficient of 0.909 to predict EBV status from gastric cancer WSI, which is expected to reduce clinical costs

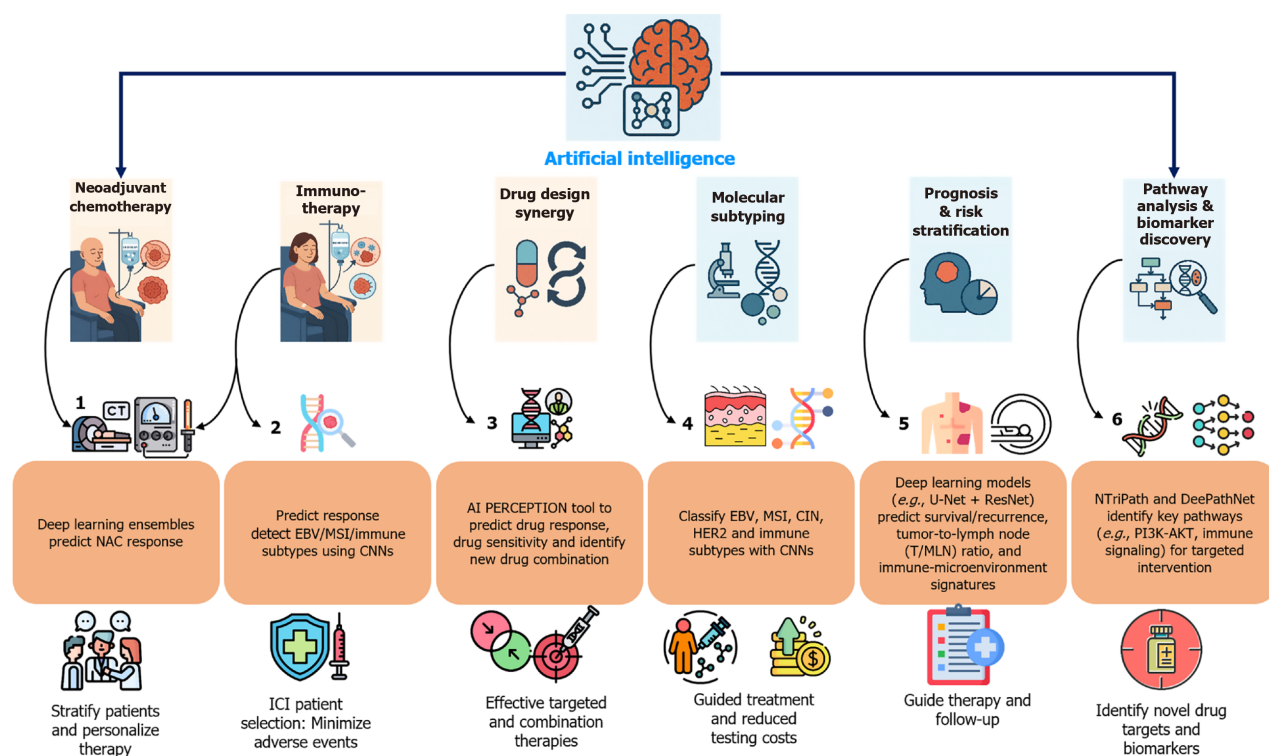


Figure 3 Artificial intelligence applications in gastrointestinal cancer therapeutics. This schematic illustrates artificial intelligence approaches for neoadjuvant chemotherapy optimization and precision oncology. Panel labels (1-6) denote data sources: (1) Computed tomography (CT) scans + radiomics; (2) Radiomics + gene expression; (3) Multi-omics (genomics of drug sensitivity in cancer/cancer cell line encyclopedia); (4) Histology + genomics; (5) Whole slide imaging, CT, and immunohistochemistry; and (6) Somatic mutations + networks. NAC: Neoadjuvant chemotherapy; EBV: Epstein-Barr virus; MSI: Microsatellite instability; CNN: Convolutional neural network; CIN: Chromosomal instability; T/MLN: Tumor/metastasis lymph node; ICI: Immune checkpoint inhibitor.

and tissue waste.

Human epidermal growth factor receptor 2 (HER2) plays a crucial role in the poor prognosis and pathogenesis of various cancers. In the case of advanced HER2-positive gastric cancer[95]. “Trastuzumab + chemotherapy” is the primary treatment option. The DESTINY-Gastric01 trial revealed that trastuzumab deruxtecan led to significant clinical improvements in patients with HER2-positive gastric cancer[96]. He *et al*[97] developed and validated a DL model using CT images of patients undergoing anti-HER2 targeted therapy. The model, incorporating multifocal and time series data, achieved one-year AUCs of 0.894 and 0.809 for Nomo-LDLM-2F using time-series medical images and tumor markers, respectively. Chen *et al*[98] demonstrated that (18)F-fludeoxyglucose positron emission tomography/CT scans could potentially aid in predicting HER2 status and guiding treatment decisions for gastric cancer[99].

Drug design and discovery

AI algorithms can predict compound efficacy and toxicity while optimizing potential drug candidates by analyzing multi-omics datasets[100]. This capability addresses the high failure rate of conventional approaches in developing treatment for aggressive GI cancers like pancreatic and advanced gastric cancers. AI-powered analysis of drug sensitization data from the genomics of drug sensitivity in cancer database, researchers identified UMI-77 as a promising therapeutic compound for specific gastric cancer subtypes[68]. The efficacy of UMI-77 correlated directly with genomic mutation signatures, with the AGS cell line (classified as high-risk) showing greater sensitivity than the lower-risk MKN45 cell line.

Prediction of drug efficacy and synergy

A ML model achieved 92% accuracy in predicting how individual cells respond to both single drugs and combinations of drugs, using single-cell RNA sequencing data from 50000 cells across 10 cancer types[101]. In a multicenter study involving 2500 patients, AI algorithms trained to predict immunotherapy response showed an average accuracy of 89% in identifying responders *vs* non-responders across melanoma, lung cancer, and gastroesophageal cancers[102]. The integration of AI with multi-omics data has led to the identification of novel drug combinations for personalized cancer treatment. A novel AI network developed a GMS that reliably predicts survival benefit for GI cancer patients receiving ICI therapy across multiple validation cohorts[68]. The GMS outperformed conventional clinical and molecular features in predicting 6, 12 and 24-month overall survival with high sensitivity and specificity. Moreover, low-risk samples identified by this AI models showed higher presence of cytolytic immune cells and enhanced immunogenic potential, explaining their superior response to immunotherapy. Some of the important works on evolving techniques of artificial intelligence for diagnosis of gastrointestinal cancer with accuracy are cited in Table 1.

Molecular subtyping for precision medicine

Mayo clinic researchers leveraged AI to develop a 32-gene signature that functions as both a prognostic and predictive biomarker for gastric cancer patients[103]. Their NTriPath ML algorithm integrated somatic mutation data with gene-to-gene interaction networks to identify cancer-associated molecular pathways with prognostic significance. The quadratic phenotypic optimization platform has demonstrated significant improvements in designing patient-specific drug combinations. CTO 2.0, a large-population AI SaaS solution, has been employed in the design of over 100 clinical trials in oncology and hematology[104]. AI tools for predicting response to ICIs have also shown promise in recent studies[105].

CLINICAL IMPLEMENTATION

AI has changed the face of medical imaging, enhancing diagnostic accuracy, streamlining workflows, and improving patient outcomes[106]. Recent years have witnessed the integration of sophisticated AI technologies-particularly DL algorithms, CNN, and generative adversarial networks-into medical imaging systems[107]. AI algorithms can analyze medical images at a scale and depth unattainable by manual review. They consistently identify subtle anomalies-such as early-stage tumors or microcalcifications in mammography-that might be missed by the human eye, thereby improving sensitivity and specificity in diagnoses[108]. As AI continues to evolve in GI cancer care, we can expect further refinements in therapeutic prediction models and novel drug discovery approaches specifically tailored to the unique molecular landscapes of different GI malignancies.

Radiomics and feature extraction

Radiomics, the extraction of quantifiable features from medical images, has emerged as a powerful tool in AI-driven medical imaging analysis. Standardized radiomic features, as approved by the Image Biomarker Standardization Initiative, can now be extracted using freely available Python scripts like PyRadiomics. Radiomic features are categorized based on their calculation methods, including intensity-based statistical, morphological, and texture matrix based features [109]. Feature reduction techniques are crucial in model building, as thousands of radiomic features can be extracted from a single image[110]. Methods such as principal component analysis and linear discriminant analysis are commonly employed. Recent studies have explored the use of CNN for features can be extraction in radiomics, potentially offering more robust and adaptable feature sets[111]. A multiparametric MRI-based radiomics model demonstrated strong noninvasive predictive performance for programmed cell death ligand 2 expression in hepatocellular carcinoma with the combined-sequence model achieving an AUC of 0.871 offering potential guidance for immune checkpoint blockade therapy selection[112].

Computer-aided detection and diagnosis

Along with radiomics, computer-aided detection and diagnosis (CAD) systems have significantly advanced, integrating various AI techniques to assist radiologists in their decision-making processes[113]. CAD applications span multiple imaging modalities, including radiography, CT, ultrasound, MRI, and radionuclide imaging, across various anatomic systems. These systems not only detect abnormalities but also assist in lesion classification, disease quantification, cancer risk assessment, and physiologic evaluation. AI algorithms are expected to not only highlight abnormalities but also suggest potential diagnoses and probabilities, significantly improving a radiologist's confidence. Natural language processing (NLP) is being integrated into CAD systems to streamline reporting processes, automatically generating structured reports based on AI-identified findings[114].

Image segmentation and tumor delineation

Accurate delineation of tumor boundaries is critical for diagnosis, treatment planning, and monitoring of therapeutic response in oncology[115]. AI, particularly DL approaches, has automated, objective and reproducible measurements that were previously reliant on time-consuming manual processes. The application of AI in tumor segmentation spans multiple imaging modalities and cancer types. Deep neural networks, including CNNs and specialized architectures like U-Net, have demonstrated remarkable capability in extracting hierarchical features and capturing complex patterns in medical images without the need for manual feature engineering[116]. These methods have significant advantages over traditional segmentation approaches such as thresholding, region-based, or edge-based methods, particularly when dealing with heterogeneous tumors that may be difficult to delineate manually. MRI segmentation has benefited substantially from DL techniques[117].

Various approaches have been developed, including models using gaussian mixture models, principal component analysis, and wavelet transforms to extract discriminative features for distinguishing tumor areas from normal tissue. In RO, AI-assisted delineation has also shown promise. Intensity-modulated RT required differentiation between tumor targets and surrounding organs at risk[118]. While manual delineation remains the standard approach, it is time-consuming, tedious, and subject to significant interobserver variation. Recent studies across multiple cancer types have evaluated AI-based segmentation tools. AI-auto-segmentation methods based on deep dilated CNNs have achieved significantly higher accuracy compared to both unmodified AI auto-contouring and fully manual delineation by radiation oncologists in lung cancer[119]. On the other hand, AI techniques reduced segmentation time from 30 minutes (manual) to just one minute while maintaining good performance across multiple structures[120]. Evolution metrics commonly used to assess segmentation performance include dice similarity coefficient, Harsdorf distance, and mean distance of agreement. These metrics provide quantitative measures of overlap and boundary accuracy between AI-generated and

ground truth segmentations[15].

Monitoring treatment response

The integration of AI in monitoring treatment response has transformed the assessment of therapeutic efficacy, disease progression, and patient outcomes in oncology. DL approaches utilizing time-series imaging have emerged as powerful tools for predicting cancer-specific outcomes, including survival, disease progression, distant metastases, and local response[121]. CNNs and recurrent neural networks have been successfully applied to predict survival and clinical endpoints in non-small cell lung cancer patients by analyzing both pre-treatment and follow-up CT images. A significant advantage of these models is their usability-requiring only single-click seed points without the need for complex volumetric segmentations, facilitating the incorporation of large numbers of scans from multiple time points into DL analyses. CT imaging-based predictions have direct applications in response assessment for clinical trials, precision medicine practices, and development of tailored clinical therapies[122]. Beyond conventional imaging, AI systems are being developed for real-time monitoring of cancer patients. These systems analyze data from multiple sources including wearing devices, electronic health records, and patient-reported outcomes (PROs) to facilitate early detection of treatment-related complications and predict disease progression. While still primarily in the research phase, these technologies show promise for enabling timely interventions and dynamic adjustment of treatment strategies. AI-based prognostic models represent another frontier in treatment monitoring, using comprehensive patient data to predict disease outcomes and survival rates with enhanced accuracy. These algorithms can potentially develop more personalized treatment plans and supportive care strategies by synthesizing numerous variables from clinical, genetic and imaging data[123]. AI combined with next-generation sequencing can identify new therapeutic targets, evaluate sensitivity and resistance to anticancer drugs, and monitor tumor evolution[124].

CELLULAR AND MOLECULAR IMAGING ANALYSIS

AI technologies have significantly advanced our ability to analyze complex cellular and molecular imaging data in cancer research, enabling unprecedented insights into tumor biology at multiple scales. Certain AI-powered tools are now able to decode complex cellular behaviors and genetic drivers directly from imaging data, accelerating both research and clinical decision-making, by bridging high molecular imaging with molecular profiling[103]. AI synthesizes data from diverse sources-radiology, pathology, genomics-to create unified models of tumor behavior. ResNet-based architectures extract morphological features from ultrasound images to classify soft tissue tumors, while U-Net models analyze cone beam CT scans for rectal cancer with 85% accuracy compared to positional CT[125]. These integrations enable predictive frameworks that correlate imaging findings with molecular targets, such as EGFR mutations or programmed cell death ligand 1 expression[126]. Also, DL nomograms differentiate between benign and malignant growths using radiomic features from CT scans, while generative adversarial networks simulate drug responses in virtual cell populations[127]. Some applications of AI in diagnosis of gastrointestinal cancer are discussed in Table 1.

Single-cell analysis and profiling

DL frameworks and AI applied to single-cell data can unravel complex gene transcription profiles and mutation spectra within tumors, helping identify sub-clonal populations that drive disease progression and metastasis[128]. AI-driven approaches for single-cell analysis in cancer include numerous frameworks with distinct capabilities. scDEAL, a deep transfer learning framework integrating bulk and single-cell RNA sequencing data, utilizes domain-adaptive neural networks to predict single-cell drug responses without depending on predefined cell labels[129]. This model can further predict critical genes that significantly contribute to drug sensitivity and resistance prediction, providing biological interpretability alongside predictive power. Another notable approach is scTherapy, which leverages single-cell transcriptomic profiles to prioritize multi-targeting treatment options for individual patients with hematological cancers or solid tumors[130]. This ML model identifies patient-specific treatments by analyzing transcriptomic differences between genetically distinct cancer cell populations compared to non-cancerous cells within the same sample. Experimental validations confirm that 96% of the multi-targeting treatments exhibit selective efficacy or synergy, and 83% demonstrate low toxicity to normal cells, highlighting their potential for therapeutic efficacy and safety. CNN-based models have been designed to predict antitumor drugs for circulating tumor cells at the single-cell level, with analysis of single-cell subsets identifying combination therapies that target mutually exclusive pathways, proving more effective than monotherapy in patient-derived xenograft models[131].

Spatial transcriptomics and proteomics

Spatial profiling technologies have transformed our understanding of the spatial organization and molecular interactions within tumors. Unlike traditional bulk sequencing, these technologies provide spatial information while generating high-dimensional molecular data. A recent study demonstrated that AI-guided spatial transcriptomic analysis significantly improves recognition of biologic features relevant to patient outcomes in high-grade serous carcinoma of the ovary[132]. Researchers found that transcriptomic profiles of tumor regions identified by AI as highly associated with outcome status were more distinct than background regions from the same tumors, superior in predicting outcome, and differed in several pathways associated with chemoresistance. Alongside, DL models have enhanced spatial resolution in transcriptomics data, improving gene expression pattern analysis in complex tissues[133]. Various architectures have been developed for spatial protein profiles, which can identify spatial motifs associated with cancer recurrence and patient survival outperforming traditional models. Reference-based spatial pathology is a notable DL framework that accurately

infers and visualizes tissue architecture from spatially resolved transcriptomics, aiding in the identification of pathological features in cancer[134]. Similarly, DeepST provides accurate and scalable spatial domain identification in spatial transcriptomics data, outperforming existing methods for defining functionally distinct regions within tumor tissues[135].

Identifying and quantifying biomarkers

As mentioned earlier, personalized single-cell expression-based planning for treatments in oncology, developed by a National Institutes of Health-led team, analyzes tumor RNA in addition to DNA to fine-tune predictions of treatment responses from bulk RNA data by focusing on single cells. DL approaches have proven to accelerate biomarker discovery by analyzing spatial gene expression patterns and tumor heterogeneity[128]. Graph neural networks can capture gene expression states from whole slide images, linking histological phenotypes with gene expression patterns. In hepatocellular carcinoma, single-cell sequencing and spatial transcriptomics have uncovered the spatial distribution and functional states of tumor-infiltrating lymphocytes, identifying stimulatory dendritic cells and macrophages as potential biomarkers for predicting therapy responses[136].

Predicting chromatin interactions in cancer

Chromatin structure and organization play crucial roles in gene regulation and cancer development[137]. AI approaches are increasingly applied to predict chromatin interactions and their dysregulation in cancer, providing insights into epigenetic mechanisms underlying oncogenesis. Multiple DL architectures combine different neural network components to model complex chromatin interactions[138]. Transformer models with convolutional and graph-node co-embedding can predict gene expressions related to chromatin structure, operating with single spot image input while maintaining interpretability and enhancing accuracy[139]. These models demonstrate superior performance in spatial gene expression prediction and have the capacity to link pathology image phenotypes to gene expression regulation.

Histone modifications (chromatin immunoprecipitation-sequencing), and transcriptomics (RNA-sequencing), these models can identify regulatory elements and their interactions in cancer genomics by integrating multi-omics data including chromatin accessibility (assay for transposase accessible chromatin with high-throughput sequencing)[34]. Single-cell approaches to chromatin analysis are also emerging, with AI methods helping to interpret patterns of chromatin accessibility across different cancer cell populations, revealing how chromatin states influence gene expression programs in heterogeneous tumors and potentially identifying new therapeutic vulnerabilities.

FOUNDATIONAL MODELS FOR AI IN CANCER RESEARCH

The emergence of foundational models represents a paradigm shift in AI signaling applications for cancer research[140]. These large-scale pre-trained models serve as versatile knowledge bases that can be adapted to diverse oncological tasks, revolutionizing how researchers extract insights from complex biomedical data. The development of increasingly sophisticated foundational models has enabled unprecedented progress in integrating knowledge across multiple domains of cancer research. Large language models (LLMs) have emerged as powerful tools for mining insights from the vast corpus of oncology literature and clinical documentation[141]. These models leverage NLP capabilities to extract structured information from unstructured text, offering significant potential for accelerating cancer research and clinical decision-making.

Bidirectional encoder representations from transformers and its variants, alongside conversational models like Chat-GPT, represent the most commonly employed LLMs for data extraction from clinical text in oncology[142]. A recent scoping review found that 75% of studies utilized bidirectional encoder representations from transformers-based models, while 25% employed Chat-GPT for oncology applications. These models have been particularly effective in pan-cancer settings (46%), with specific applications in breast (17%) and lung (17%) cancer contexts[143]. Recent trends have indicated an evolution in LLM implementation strategies. Information extraction and text classification have emerged the predominant NLP tasks in cancer research. Specific applications include extracting tumor characteristics from radiology reports, identifying programmed cell death ligand 1 expression levels from clinical notes, and classifying cancer stages and metastasis presence[144]. These automated extraction capabilities significantly reduce the administrative burden of reviewing patient health records, potentially increasing time available for patient-facing care.

EMERGING APPLICATIONS OF AI IN ONCOLOGY

AI-powered tools are revolutionizing GI cancer screening and diagnosis. The GI genius system, now deployed in clinical practice, served as an AI-assisted “second set of eyes” during colonoscopy, identifying precancerous polyps in real time even those difficult to detect due to size or location[145]. This technology has significantly increased polyp detection rates, supporting earlier intervention and improved patient outcomes in CRC[146]. Similarly, AI models such as the GRAIDS have achieved diagnostic accuracies exceeding 95% in upper GI cancers, outperforming standard clinical tests and enabling real-time, high-precision detection of early gastric cancer[147]. In pathology, AI algorithm now match expert pathologists in detecting neoplastic lesions in upper endoscopy, with AUC values up to 0.96 for neoplastic lesions detection in stomach, Barrett’s esophagus and squamous esophagus[148]. As AI technologies mature, their applications extend beyond traditional areas such as diagnostics and treatment planning to encompass broader aspects of cancer

research and patient care. These emerging applications hold significant potential for improving cancer prevention, early detection, patient engagement, and the overall efficiency of healthcare delivery. Some of the important works on evolving techniques of artificial intelligence and their applications in gastrointestinal cancer management are cited in [Table 1](#).

AI in cancer epidemiology and population health

AI is increasingly being utilized to analyze large-scale population health data, offering new insights into cancer epidemiology and risk factors. ML algorithms can identify complex patterns in demographic, environmental, and lifestyle data that correlate with cancer incidence and mortality rates. AI models can identify geographic hotspots with elevated cancer risks by integrating publicly available datasets with electronic health records and social media data, enabling targeted interventions and resource allocation[149]. Such approaches can also identify high-risk populations based on factors such as smoking habits, dietary patterns, and exposure to environmental toxins, thereby facilitating personalized prevention strategies. Furthermore, AI can enhance cancer surveillance programs by automating the extraction of relevant information from unstructured data sources such as pathology reports and death certificates. Automated data extraction accelerates the identification of emerging cancer trends, facilitating rapid responses to public health challenges.

Real-world data analysis and PROs

Real world data (RWD) and PROs are becoming increasingly valuable in oncology research, providing insights into treatment effectiveness and patient experiences outside of clinical trial settings[150]. AI plays a crucial role in analyzing these complex datasets to generate actionable knowledge. NLP techniques enable automated extraction of relevant information from unstructured clinical notes, imaging reports, and other RWD sources, reducing the manual effort required for data curation[151]. ML algorithms can identify predictive biomarkers for treatment response and survival outcomes based on RWD, complementing findings from traditional clinical trials[102]. The analysis of PROs, such as symptom burden, quality of life, and functional status, provides a patient-centered perspective on cancer care. AI models can identify patterns in PRO data that correlate with treatment outcomes, enabling clinicians to tailor treatment plans based on individual patient needs and preferences[51].

AI-powered virtual assistants for patient care and education

AI-powered virtual assistants are emerging as valuable tools for patient engagement and education, providing personalized support and information throughout the cancer journey. These virtual assistants can address patient questions, provide treatment reminders, and offer emotional support, improving adherence to treatment plans and overall patient satisfaction. Chatbots powered by NLP can provide 24/7 access to reliable cancer information, addressing common patient concerns and reducing the burden on healthcare providers[152]. Virtual assistants can also facilitate remote monitoring of patient symptoms and side effects, enabling timely interventions and preventing unnecessary hospitalizations[153]. Furthermore, AI-powered virtual assistants can deliver personalized educational materials tailored to individual patient needs, improving understanding of treatment options, potential side effects, and self-management strategies.

Precision oncology and personalized treatment planning

AI is central to the advancement of precision oncology, enabling personalized treatment planning based on individual patient characteristics, tumor genomics, and other relevant factors. AI models can predict response, identify optimal drug combinations, and minimize adverse effects by integrating diverse data types[154]. ML algorithms can analyze genomic data to identify actionable mutations and predict sensitivity or resistance to targeted therapies. AI-driven approaches can also integrate radiomic features extracted from medical images to predict treatment outcomes and guide radiation therapy planning. AI can assist in the selection of optimal treatment sequences and drug combinations by simulating treatment scenarios based on individual patient data, thereby empowering clinicians to make more informed treatment decisions, improving patient outcomes and reducing healthcare costs[155].

CHALLENGES AND LIMITATIONS

Despite significant advancements in AI applications for oncology, several challenges and limitations must be addressed to ensure the responsible and effective implementation of these technologies in clinical practice.

Ethical, legal, and social implications of AI in GI oncology

The deployment of AI systems in GI cancer detection and diagnosis raises several ethical, legal, and social challenges that must be addressed to ensure responsible adoption. Bias mitigation is critical, as models trained on non-representative datasets risk perpetuating healthcare disparities particularly across underrepresented populations in screening programs. Ensuring data privacy and security is equally important, with compliance to frameworks such as Health Insurance Portability and Accountability Act in the United States and General Data Protection Regulation in the European Union governing patient data storage, sharing, and de-identification. From a clinical perspective, informed consent should extend beyond traditional procedures to include patient awareness of AI-assisted decision-making and its limitations. This is particularly relevant when AI outputs are used in risk stratification or treatment selection, where misinterpretation could have significant consequences. The regulatory landscape for AI-based medical tools is evolving, with agencies such as the United States Food and Drug Administration and the European Medicines Agency issuing guidance for algorithm

transparency, performance validation, and post-deployment monitoring. In GI oncology, ongoing validation across multi-center, multi-ethnic cohorts will be essential to meet these regulatory standards. Finally, the explainability of AI models remains a key social concern; black-box algorithms may erode trust among clinicians and patients. Incorporating explainable AI techniques could improve transparency, facilitate clinical acceptance, and ultimately support equitable, evidence-based cancer care.

Data quality and standardization

Data quality and standardization represent a critical challenge for AI applications in oncology[156]. AI models rely on large, high-quantity datasets for training and validation, but the availability of such data is often limited by inconsistencies in data collection, annotation and storage practices[157]. The lack of standardized data formats and terminologies across different healthcare institutions hinders the integration of data from multiple sources, limiting the generalizability of AI models. Efforts to standardize data collection and annotation practices are essential to improve the quality and availability of data for AI applications in oncology. Data biases can inadvertently be introduced during data collection and annotation, leading to AI models that perform poorly on certain patient populations or exhibit unintended discriminatory behavior[158].

Model generalizability and external validation

Model generalizability and external validation are essential to ensure AI models perform reliably in diverse clinical settings. Many AI models developed for oncology applications are trained and validated on data from specific institutions or patient populations, limiting their applicability to other settings. External validation, where AI models are tested on independent datasets from different sources, is crucial for assessing their generalizability and identifying potential biases[159]. Rigorous external validation studies are needed to ensure that AI models can be reliably deployed in real-world clinical practice. AI models should be continuously monitored and updated as new data becomes available to maintain their accuracy and relevance over time. Adaptive learning techniques can be used to incorporate new data and improve model performance in response to changing clinical environments[160,161].

Integration into clinical workflows

Integrating AI models into existing clinical workflows represent a significant challenge for widespread adoption of these technologies in oncology[162]. AI models must be seamlessly integrated into the clinical decision-making process to provide timely and actionable insights to healthcare providers. AI models should be designed to complement, rather than replace, human expertise. Clinicians should retain the ability to override AI recommendations based on their clinical judgement and patient-specific factors. In alignment with the clinical workflow, mandatory workforce training and adaptation are crucial for a complete and successful implementation of AI technologies in oncology[163]. Healthcare professionals need to be trained on how to effectively use and interpret AI-driven insights to make informed clinical decisions. Training programs should focus on developing AI literacy among healthcare professionals, including an understanding of the strengths and limitations of AI models, as well as the ethical considerations associated with their use.

FUTURE DIRECTIONS OF AI AND GI CANCER RESEARCH

Gastric cancer is one of the most common malignant tumors with high mortality. Multiple studies have explored the role and applications of AI on multiple tasks in the diagnosis and treatment of GI cancers. AI can help detect precancerous lesions and early tumors in endoscopic examinations, thereby reducing the time for diagnosis and do multiple WSI image subtyping thereafter[164]. It could also assist with TNM staging, treatment selection and prognosis prediction for patients with advanced gastric cancer based on CT images. Having said that, current approaches based on AI in GI cancer diagnosis are full of challenges. The data scarcity and poor interpretability could be improved majorly by data regularization and advanced algorithms. As AI technologies mature, their impact will extend from accelerating drug discovery and enhancing diagnostic precision to enabling personalized treatment strategies and improving patient outcomes on a global scale[165]. The next phase of AI in cancer research will be defined by several key trends: The adoption of quantum computing to solve complex biological problems, fostering effective AI-human collaboration in clinical decision-making, integrating AI seamlessly across all stages of the cancer care continuum, and establishing robust global collaborations and data sharing initiatives[166].

Quantum computing in cancer research

The integration of quantum computing with AI represents a revolutionary frontier in GI cancer research, offering unprecedented computational power to address complex oncological challenges. Recent collaborative research between University of Toronto and Insilico Medicine demonstrated the potential of quantum computing and AI to transform drug discovery by creating molecules targeting *KRAS*, a cancer-driving protein previously considered "undruggable"[167]. This quantum-based approach enables precise modeling of drug-target interactions, allowing researchers to efficiently screen millions of small-molecule candidates and optimize existing drugs. Other hybrid quantum-classical algorithms have successfully identified novel inhibitors for *KRAS* mutations, which are present in approximately 25% of human cancers[167]. Beyond drug discovery, quantum technologies offer substantial benefits for cancer diagnostics through enhanced medical imaging capabilities. Quantum algorithms can generate higher-definition magnetic resonance images and detect finer features, while quantum ML models can analyze imaging data to uncover subtle diagnostic and

prognostic signatures. These advancements allow for improved resolution and enable new imaging capabilities that could revolutionize early cancer detection. Quantum sensing represents another promising application, with quantum dots offering high sensitivity and selectivity for non-invasive measurement of biological tissues. These sensors have the potential to target specific macromolecules and detect minute cellular and metabolic changes, potentially enabling earlier cancer detection and personalized treatment planning based on continuous monitoring[168].

AI-human collaboration in oncology

The future of oncology increasingly depends on effective collaboration between AI systems and healthcare professionals. This partnership leverages the computational power and pattern recognition capabilities of AI while preserving the clinical expertise, ethical judgment, and human compassion essential to cancer care[169]. Research on human-AI interaction in dynamic decision-making for cancer treatment reveals that this collaborative process is multifactorial, influenced by the complex interrelationship between prior knowledge and preferences, patient state, disease site, treatment modality, model transparency, and AI's learned behavior. Studies show various clinician responses to AI recommendations: Some may disregard them due to skepticism; Others critically analyze them case-by-case; Most adjust their decisions when AI recommendations benefit patients but disregard them when deemed harmful or suboptimal[170]. In diagnostic contexts, particularly prostate cancer diagnosis, human-AI collaboration has demonstrated significant potential for reducing variability and improving workflow efficiency[171]. While AI systems can sometimes outperform single radiologists, they do not yet exceed the accuracy of expert radiologists working within multidisciplinary teams or match the accuracy of two independent readers in screening scenarios. This highlights the continued importance of human expertise in interpreting and contextualizing AI outputs. LLMs show promise for treatment decision support in oncology, though there are ongoing challenges regarding accuracy and reliability. Recent studies evaluating LLMs in suggesting treatment options found they generated a wider range of options compared to human experts, though their recommendations sometimes deviated from expert consensus[172]. The integration of these AI tools must be approached thoughtfully, ensuring they complement rather than replace the human expertise and compassion that define oncology practice.

Integrating AI across the cancer care continuum

AI is increasingly being integrated throughout the entire cancer care continuum, from prevention and screening to diagnosis, treatment, and survivorship. This comprehensive approach maximizes the impact of AI on patient outcomes while addressing various challenges at each stage of cancer management. In early detection and diagnosis, AI algorithms are enhancing the accuracy of cancer screening by identifying subtle patterns in imaging data that may be imperceptible to human observers[170]. These applications are maturing beyond research and development to direct clinical integration across various cancer types and data modalities, including imaging, genomics, and medical records. AI enables more personalized approaches by analyzing complex medical data including genetic profiles, patient histories, and treatment responses. Predictive analytics powered by AI help clinicians anticipate how patients will respond to different therapies, enabling them to make more informed decisions about treatment plans, adjust therapies as needed, and better manage patient expectations. Real-time monitoring of treatment response represents another promising application, with AI systems analyzing data from multiple sources including wearable devices, electronic health records, and PROs[171].

CONCLUSION

The future of AI in cancer research increasingly depends on large-scale collaborations and data sharing initiatives that overcome institutional and geographic boundaries to accelerate discovery and translation. A prominent example is the cancer AI alliance (CAIA), a collaboration between four National Cancer Institute-designated cancer centers with funding from AWS, Deloitte, Microsoft, and NVIDIA. This alliance aims to apply responsible AI to transform cancer research and care by unlocking insights while protecting data security. The initiative will provide shared infrastructure and shape industry standards, supporting greater health outcomes by exposing data trends for rare cancers and small populations. Critical to these collaborations is maintaining data security and privacy. CAIA employs a federated AI learning framework where each cancer center maintains independent data while AI models are sent to the data to produce results. These results are then aggregated across participating members to uncover insights without sharing or exposing raw data. International collaborations are also advancing, exemplified by the lung project, an European-backed initiative aiming to individualize treatment in patients with advanced non-small-cell lung cancer[172]. This initiative enables data harmonization across partner institutions including six European Union cancer centers and centers in the United States and Israel, supporting model refinement and cross-institutional omics data standardization. Public-private partnerships further accelerate progress, as demonstrated by Lunit's collaboration with the National Cancer Institute to advance AI-powered biomarker research. Under this agreement, Lunit tools are being made available to National Cancer Institute Center for Cancer Research investigators across their clinical trials portfolio, applying AI technologies to analyze whole-slide images and develop data-driven insights for personalizing cancer treatment approaches. These collaborative initiatives collectively promise to overcome key challenges in AI adoption for oncology, including data standardization, model validation, and equitable access to advanced technologies-ultimately accelerating the development of more effective and personalized cancer care worldwide.

FOOTNOTES

Author contributions: Verma HK lead the study; Suri C, Ratre YK were involved in the data collection and validation, provided the first draft of the manuscript; Ratre YK and Pande B prepared the figures and tables; Suri C, Ratre YK, Verma HK and Bhaskar L wrote and finalized the manuscript; Verma HK and Bhaskar L designed the outline and coordinated the writing of the paper; all authors have read and agreed to the published version of the manuscript.

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